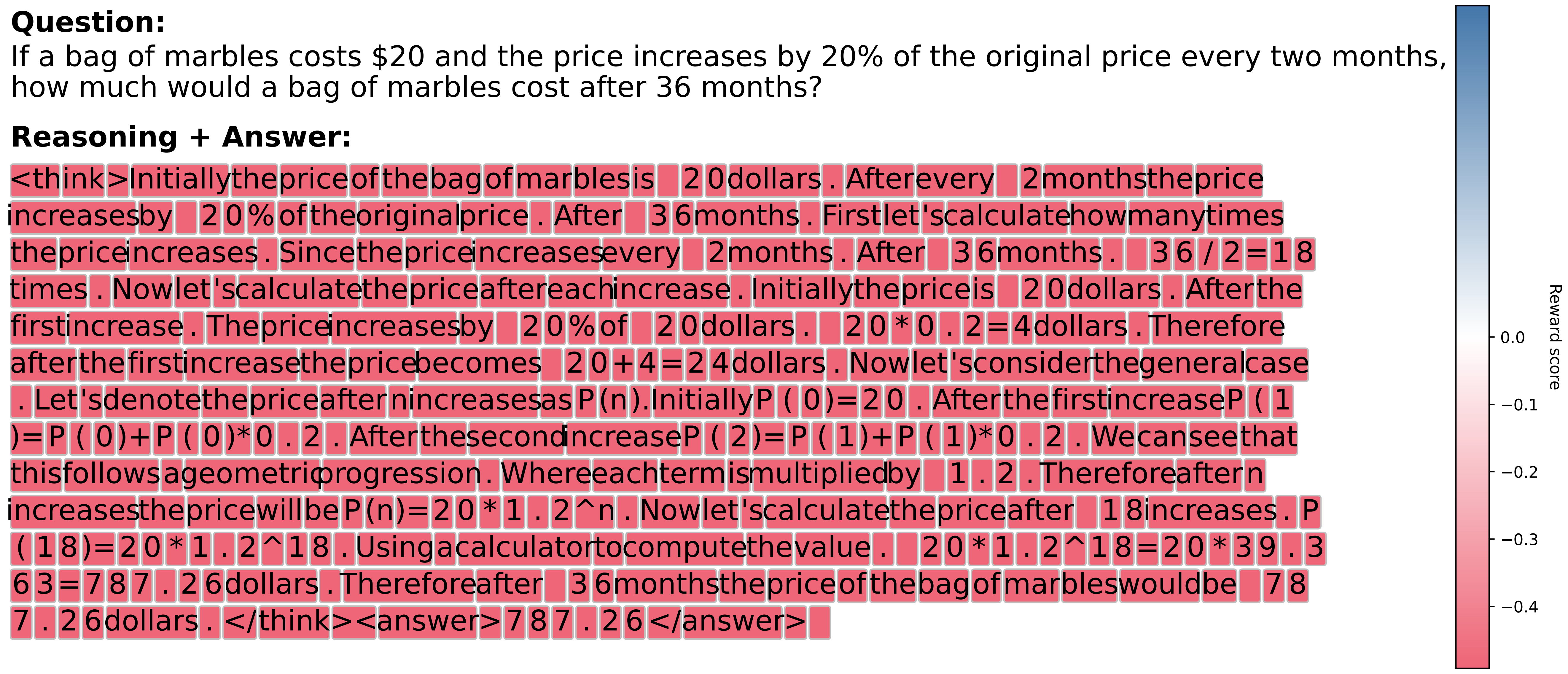
Question:
If a bag of marbles costs $20 and the price increases by 20% of the original price every two months, how much would a bag of marbles cost after 36 months?
Reasoning + Answer:
<think>Initially the price of the bag of marbles is 20 dollars. After every 2 months the price increases by 20% of the original price. After 36 months. First let's calculate how many times the price increases. Since the price increases every 2 months. After 36 months. 36/2=18 times. Now let's calculate the price after each increase. Initially the price is 20 dollars. After the first increase. The price increases by 20% of 20 dollars. 20*0.2=4 dollars. Therefore after the first increase the price becomes 20+4=24 dollars. Now let's consider the general case. Let's denote the price after n increases as P(n). Initially P(0)=20. After the first increase P(1)=P(0)+P(0)*0.2. After the second increase P(2)=P(1)+P(1)*0.2. We can see that this follows a geometric progression. Where each term is multiplied by 1.2. Therefore after n increases the price will be P(n)=20*1.2^n. Now let's calculate the price after 18 increases. P(18)=20*1.2^18. Using a calculator to compute the value. 20*1.2^18=20*39.363=787.26 dollars. Therefore after 36 months the price of the bag of marbles would be 787.26 dollars.</think><answer>787.26</answer>
0.0
−0.1
−0.2
−0.3
−0.4
Reward score